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Lecture : Introduction to Data Science Tools

Term : Fall 2024

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===== **Lecture 1 Notes**

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Program: Set of **instructions** for computer to perform a task.

$$y = 2 * x - 5 / 4 * z - w$$

Compiler: It is a program that converts a code (written in any High-Level Programming Language, HLPL) into machine language (binary language).

Computer? An electronic device that is capable of doing four main tasks: Input, process (ALU stuff), storage, and output.

Computers are STUPID but FAST, human being on the other hand SMART but SLOW (Einstein).

Construct of a structured HLPL:

- Variable
- Constants
- Tokens (i, =, !, >, y, ...) → expressions (i, z>5, 2/4, x==y) → statements. (a complete expression)
- Control structures (Conditional Executions structures or Branching Constructs)
 - o If
 - o If-else
 - o Switch-case-break
- Repetitions (Loops): Doing the same thing over and over again.
 - o For
 - o While
 - o Do-while
 - o Until
- Arrays (a defined data structure to hold many values with a single name)
- Files (open, close, read, write, append)
- (User defined) Functions

Additional Constructs for **Object-Oriented** HLPL

- Classes & Objects
- Inheritance
- Polymorphisms

Question? What is the difference between **object-based** (such as JavaScript) and Object-oriented (such as Java) Programming languages?

Registers:

- a) Address
- b) Data
- c) Instructions (Fetched->Decoded→Executed→Written back to memory)

Notes:

1. All programs run on memory

Exercises:

1. Write a calculator program in C that takes 3 inputs from command line and performs the necessary operation. For example

my_calc 3 + 4 → 3+4 = 7

my_calc 3 / 5 → 3/5 = 0.6000

my_calc 5 x 2 → 5*2 = 10